

Md. Dulal Hossain, Teaching Assistant

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Experience

- Oct 2025 – Present  **Teaching Assistant**, Green University of Bangladesh.
Skills: Python, Data Structures, Data Mining, Algorithms, Computer Networking, Information System and Design, Student Mentorship, Academic Grading, Research Assistance, Thesis Guidance, Documentation & Project Coordination.
- Aug 2024 – Present  **Research Assistant**, Professor Dr. Md. Ahsan Habib
Excited to coordinate research, manage academic tasks, & support impactful projects while learning, growing, & contributing under his inspiring mentorship in my academic journey.
- Oct 2024 – Oct 2025  **Applied Researcher**, Nixtec Systems, Bangladesh & Netherlands (Remote).
Skills: Blockchain, Machine Learning, Deep Learning, Natural Language Processing & Large Language Models.
- Dec 2024 – Apr 2025  **Internship**, Wizard Software & Technology Bangladesh Ltd, Dhaka, Bangladesh.
Skills: Automation, Artificial Intelligence, Robotic Process Automation, Data Annotation, System Design, Data Governance.
- Jun 2025 – Oct 2025  **Instructor (AI, ML, DM & DL)**, EduTock - Blended Learning Ecosystem
Conducting training sessions and workshops on Artificial Intelligence, Machine Learning, Data Mining, and Deep Learning.

Education

- Nov 2021 – Oct 2025  **B.Sc. in CSE, Green University of Bangladesh**, Dhaka, Bangladesh.
CGPA: 3.60 out of 4 (Passing Years 2025).
Thesis title: *Development of Blockchain-based Framework for Certificate Verification and Fraud Detection.*
- Jun 2017 – May 2020  **Higher Secondary School Certificate (HSC)**, Science, Moulana Saifuddin Ahiya Degree College Shahzadpur, Sirajgonj.
GPA: 4.17 out of 5 (Passing Years 2020).
- Jan 2012 – Mar 2017  **Secondary School Certificate (SSC)**, Science, Jamirta High School Jamirta, Shahzadpur, Sirajgonj.
GPA: 4.41 out of 5 (Passing Years 2017).

Research Publications

Conference Proceedings

-  **Hossain, Md. Dulal**, M. J. H. Mahin, M. R. Khan, M. Akther, and M. A. Habib, "Hybrid deep learning and ensemble methods for dependable IoT intrusion detection," in *Proceedings of the 2025 7th International Conference on Sustainable Technologies for Industry 5.0 (STI)*, Dhaka, Bangladesh: IEEE, 2025, pp. 1–6.

- 2 **Hossain, Md. Dulal** et al., "Bcvfd: Development of blockchain-based framework for certificate verification and fraud detection," in *Proceedings of the IEEE Computer Society Bangladesh Chapter Summer Symposium 2025*, HSTU, Dinajpur, Bangladesh: IEEE, 2025, p. 1.
- 3 M. N. Pathan, **Hossain, Md. Dulal**, A. Sultana, A. Hasan, S. C. Das, and M. Ashiqussalehin, "A hybrid deep ensemble framework for intelligent intrusion detection and classification in data center environments," in *Proceedings of the 2025 7th International Conference on Sustainable Technologies for Industry 5.0 (STI)*, Dhaka, Bangladesh: IEEE, 2025, pp. 1–6.

Manuscripts in Preparation for Journals

- 1 **Hossain, Md. Dulal** et al., "Bcvfd: Development of blockchain-based framework for certificate verification and fraud detection," *To be added*, Feb 2026.
- 2 M. N. Piyash, K. N. Uddin, **Hossain, Md. Dulal**, and M. A. Habib, "A privacy-preserving federated learning framework for smart healthcare using blockchain and zk-snarks," *To be added*, Nov 2025.
- 3 T. Ahammed, D. Mondal, **Hossain, Md. Dulal**, and M. A. Habib, "Parameter-efficient bangla code generation using code-specialized large language models," *To be added*, 2025.
- 4 M. Atik, M. A. R. Sarkar, M. T. Rahman, **Hossain, Md. Dulal**, and M. A. Habib, "Optimizing real-time distributed systems with event-driven architectures in edge computing environments," *To be added*, 2025.
- 5 A. A. S. Hasnat, M. R. Karim, Z. H. Sajjad, **Hossain, Md. Dulal**, and M. A. Habib, "Improvement of qoe in smart environments through mulsemmedia," *To be added*, 2025.
- 6 M. J. H. Mahin, M. J. Khan, M. A. Mia, **Hossain, Md. Dulal**, and M. A. Habib, "Nid verification system with blockchain-enabled ids," *To be added*, 2025.
- 7 M. E. Mia, M. S. Uddin, M. J. Nayeem, **Hossain, Md. Dulal**, and M. A. Habib, "Adaptive and dynamic blockchain framework for secure data sharing in zero-trust edge environments," *To be added*, 2025.
- 8 S. H. Nishan, M. Harun-Ur-Roshid, M. Akter, **Hossain, Md. Dulal**, and M. A. Habib, "Seamless 6g connectivity: Ai-based handover mechanisms for hybrid satellite-terrestrial networks," *To be added*, 2025.
- 9 A. Rahman, F. Foysal, M. Rakibul, **Hossain, Md. Dulal**, and M. A. Habib, "Joint service migration and resource allocation in cloud computing using large language models (llms)," *To be added*, 2025.
- 10 M. M. Rahman, M. H. Showrab, S. Sultana, **Hossain, Md. Dulal**, and M. A. Habib, "Development of 5g surveillance network for maritime border of bangladesh," *To be added*, 2025.
- 11 A. I. Sezan, R. Chowdhury, N. L. Khan, R. Palas, **Hossain, Md. Dulal**, and M. A. Habib, "Food supply chain transparency and efficiency through the integration of iot and blockchain," *To be added*, 2025.

Manuscripts in Preparation for Conferences

- 1 **Hossain, Md. Dulal**, M. Akther, M. R. Khan, M. J. H. Mahin, and M. A. Habib, *Revealing colorectal cancer's genetic clues through smart bioinformatics*, Manuscript in preparation for conference submission, 2025.
- 2 **Hossain, Md. Dulal**, Rakib, F. Saha, M. Akther, M. N. Pathan, and M. A. Habib, *An intelligent machine learning framework for disease prediction and personalized medical recommendations*, 2025.
- 3 A. Khatun, M. H. Lemon, N. Mollik, **Hossain, Md. Dulal**, and M. A. Habib, *Development of fish farming advisory system using ai*, 2025.
- 4 M. N. H. Riyad, K. Osama, D. Datta, **Hossain, Md. Dulal**, and M. A. Habib, *Skill-based employee recommendation through ai-based resume screening for job portals*, 2025.

Skills

Languages	📌 Bengali (Native), English (Fluent - Academic & Professional)
Coding	📌 Python, Java, JavaScript, TypeScript, C, C++, Assembly, HTML, CSS, SQL, Shell, PowerShell, Markdown & LaTeX
Frameworks	📌 Odoo, Django, React.js, Node.js, Express.js, Bootstrap, Tailwind CSS & Scikit-learn
Software Meth.	📌 Scrum, Waterfall, V-Model, Spiral Model, Iterative Model, Prototype Model & Extreme Programming (XP)
Databases	📌 MySQL, MongoDB, PostgreSQL & Oracle
Web Dev	📌 HTML, CSS, JavaScript, React, PHP, Apache Web Server
Miscellaneous	📌 Research, Teaching, Training Data Analysis, Machine Learning, Software Development Life Cycle, $\LaTeX$ typesetting and publishing

Projects

Full Stack and Web Development Projects

- Aug 2025 📌 **BanglaHaat Shopping (Full Stack E-Commerce Platform)**, Developed using React.js, Node.js, Express.js, Tailwind CSS, MongoDB, Redux Toolkit, Stripe, Cloudinary, Git, GitHub, VS Code & Vercel
A full-stack e-commerce platform enabling users to shop, manage products, and make secure online payments through an easy, responsive interface.
- May 2025 📌 **Intelligent Review Analyser**, Developed using Python
Uses SVM and BERT models to classify customer reviews as positive, negative, or neutral, achieving 79% accuracy.
- Apr 2025 📌 **PET Management System**, Developed using HTML, CSS, JavaScript, PHP, and MySQL
Web-based system to manage pet information, appointments, and medical records. Developed using PHP, MySQL, and Bootstrap.
- Jan 2025 📌 **Desktop Voice Assistant (Familiar Siri, Alexa)**, Developed using Python
Interactive system with speech recognition and text-to-speech features. Performs tasks like providing time, playing music, and accessing weather updates.
- May 2024 📌 **E-Commerce Website: SDM Dulal PC Shop**, Developed using HTML, CSS, JavaScript, PHP, and MySQL
Platform with user registration, product selection, cart management, and admin panel. Includes a unique repair service feature.

Engineering and Technical Projects

- 📌 **Group Chat Application (Familiar Whatsapp) Group Chat Application (Familiar Whatsapp)**, Developed in Java
Java-based client-server system supporting real-time messaging, image sharing, emoji integration, and an intuitive interface using JavaFX.
- Dec 2023 📌 **School Management System (Familiar GUMS)**, Using DBMS and Java
Modules for student addition, admin management, fee tracking, result processing, and staff administration. Centralized platform for school management.
- May 2023 📌 **Bridge Construction (Padma) - Engineering Drawing Lab Project**, Using AutoCAD
Designs a 150-meter bridge section using AutoCAD for safety and connectivity.
- Apr 2023 📌 **Banking System (Familiar Nogod)**, Developed using online technologies
Automates manual banking processes with balance checking, fund transfer, and transaction record maintenance.

Projects (continued)

Other Programming Projects

- Mar 2024  **Library Management System Application with Interface**, Using Shell Scripting
Bash-based command-line tool to add, find, edit, remove, and view book records efficiently.
- Oct 2023  **Application of Book Store (Familiar Retailer Apps)**, Developed using Assembly Language
Book store application demonstrating efficient inventory management and transaction processing in low-level programming.
- 2020-2025  **Additional Projects on GitHub**, Using Multiple Languages
Developed 30+ additional applications in web development, data science, machine learning, blockchain, and more.

Social Work History

- Mar 2025 – Oct 2025  **Spokesperson of GUB Transport (Lead)**, Green University of Bangladesh.
Skills: Green University of Bangladesh Transport Committee.
- Dec 2021 – Oct 2025  **Home Tutor (English Medium)**, Oxford International School Dhaka (Student)
I am a home tutor specializing in English medium education.
- Jun 2017 – May 2019  **Bangladesh Scouts (Member)**, Moulana Saifuddin Ahiya Degree College, Shahzadpur, Sirajgonj
Actively participated in leadership, teamwork, and community service initiatives. Developed organizational and interpersonal skills while contributing to college and local community events.
- Jan 2018 & Jan 2019  **Event Organizer (Leader)**, Moulana Saifuddin Ahiya Degree College
Led event organization by coordinating teams, managing logistics, and ensuring the successful execution of college programs and activities, demonstrating leadership and organizational skills.
- Mar 2025  **Keynote Speaker**, Webinar - Cyber Existence of a Graduate's Professional Skills
Delivered a keynote on enhancing graduates' digital presence via LinkedIn, GitHub, and freelancing at an exclusive GUB webinar. [Post link](#)
- Dec 2024  **Volunteer**, Cyber Security and Cloud Systems
Organized a Cyber Security seminar, ensuring smooth coordination, participant engagement, and overall event success.
- May 2025  **Volunteer (Anchor)**, Closing Ceremony GUB EDGE Program
Hosted the closing ceremony, coordinated sessions, and engaged the audience as Anchor for GUB's Enhancing Digital Government and Economy (EDGE) Program final event.
- Apr 2025  **Volunteer (Anchor)**, Unlock your AI Potential with Agent X!
Hosted the closing ceremony, coordinated sessions, and engaged the audience as Anchor for GUB's AI-focused event, ensuring smooth execution and participant engagement.

Miscellaneous Experience

Awards and Achievements

- 2023 – 2025  **Academic Excellence Awards**, Green University of Bangladesh
Consistently recognized for outstanding academic performance across three consecutive semesters: Fall 2023 (GPA 3.80), Spring 2025 (GPA 3.91), and Summer 2025 (GPA 3.90). Earned eligibility for the Dean's Certificate in all semesters for maintaining exceptional academic standards.
- Jan 2019  **Champion**, Badminton Tournament, Rajshahi
Won the championship in a competitive badminton tournament held in Rajshahi.
- Feb 2018  **Runner-up**, Badminton Tournament, Shahzadpur Upazila, Sirajgonj
Secured the runner-up position in the Shahzadpur Upazila badminton tournament, demonstrating skill and sportsmanship.

Certification

- Nov 2024  **Python & Django Developer**, Institute of Information Technology (IIT), Jahangirnagar University
Gained hands-on experience in Python programming, web development, software development, and IT infrastructure while completing the EDGE Digital Skills Development Program.
- Jun 2025  **AI+ Prompt Engineer: Level 1**, AI CERTs™ Learner Portal
Earned the AI+ Prompt Engineer™: Level 1 certification, demonstrating foundational skills in prompt engineering for AI model interaction and optimization.
-  **Prompting with Microsoft Copilot**, Microsoft
Gained hands-on experience in prompting with Microsoft Copilot, learning to generate, refine, and optimize AI-powered outputs for productivity, content creation, and task automation.
- May 2025  **Automation Anywhere (Essentials)**, Automation Anywhere
Completed the Essentials Certification covering the fundamentals of Robotic Process Automation (RPA) using Automation Anywhere.

References

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